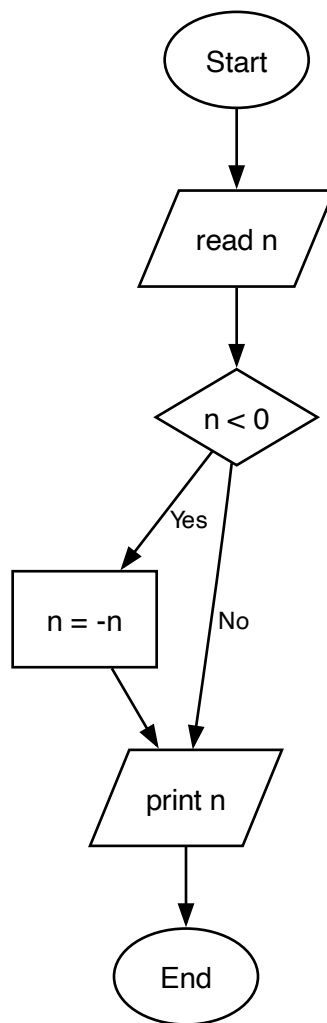


Answer key

Problem 1.



(i) For each input below, follow the flowchart and write the output.

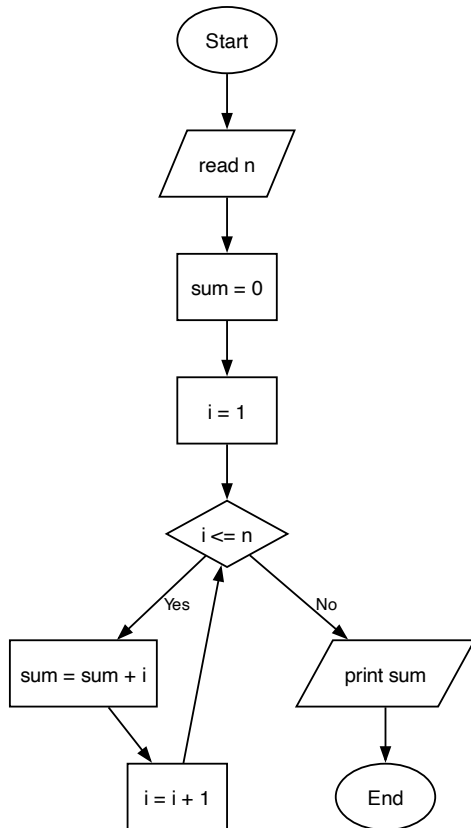
input n	output
7	7
-4	4
0	0
-19	19

(ii) Can you tell what this algorithm is doing?

Answer: Absolute value

Problem 2.

Trace this algorithm for $n = 4$, and complete the trace table.

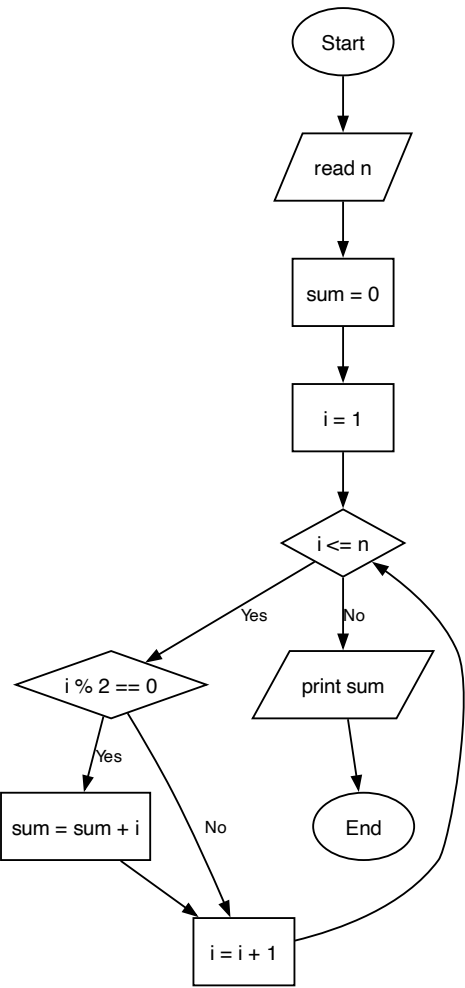


n	sum	i	output
4			
4	0		
4	0	1	
4	1	1	
4	1	2	
4	3	2	
4	3	3	
4	6	3	
4	6	4	
4	10	4	
4	10	5	
4	10	5	10

Output: 10

Problem 3.

Trace this algorithm for $n = 5$, and complete the trace table.

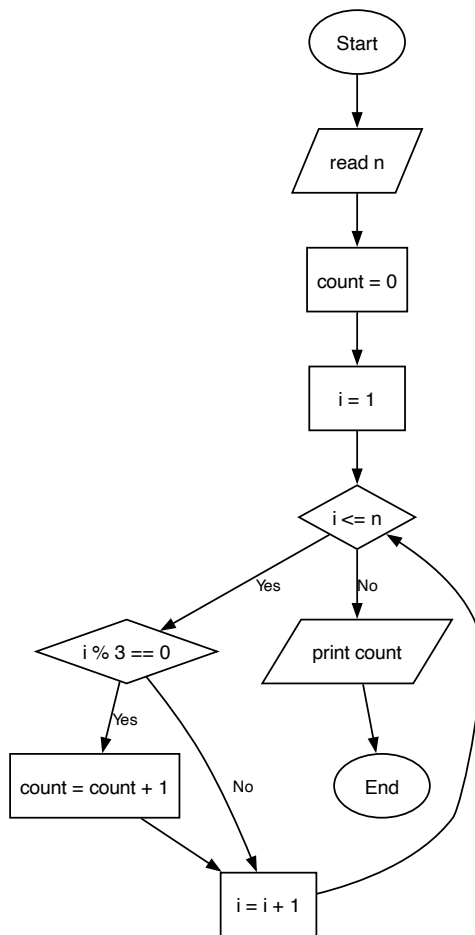


n	sum	i	output
5			
5	0		
5	0	1	
5	0	2	
5	2	2	
5	2	3	
5	2	4	
5	6	4	
5	6	5	
5	6	6	
5	6	6	6

Output: 6

Problem 4.

Trace this algorithm for $n = 7$, and complete the trace table.



n	count	i	output
7			
7	0		
7	0	1	
7	0	2	
7	0	3	
7	1	3	
7	1	4	
7	1	5	
7	1	6	
7	2	6	
7	2	7	
7	2	8	
7	2	8	2

Output: 2

Problem 5.

Trace this algorithm for $n = 472$, and complete the trace table.

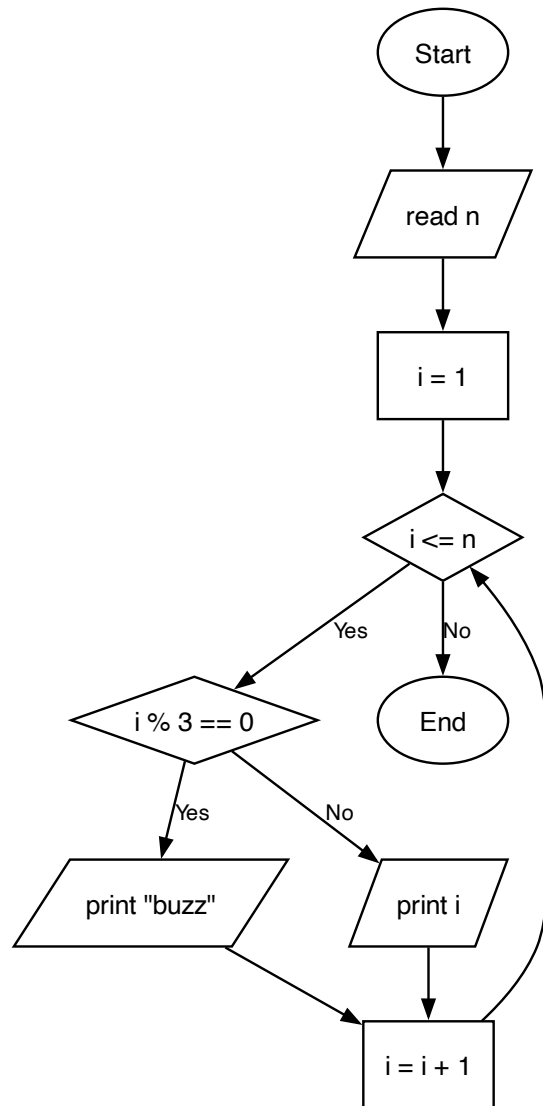
Note: $n // 10$ means whole-number division — divide and drop the remainder. For example, $472 // 10 = 47$ (not 47.2).



n	count	output
472		
472	0	
472	1	
47	1	
47	2	
4	2	
4	3	
0	3	
0	3	3

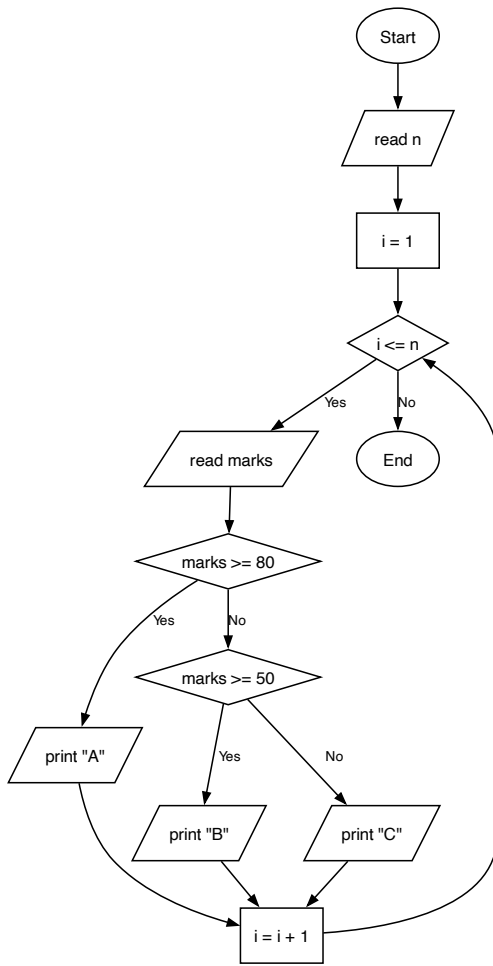
Output: 3

Problem 6.



Problem 7.

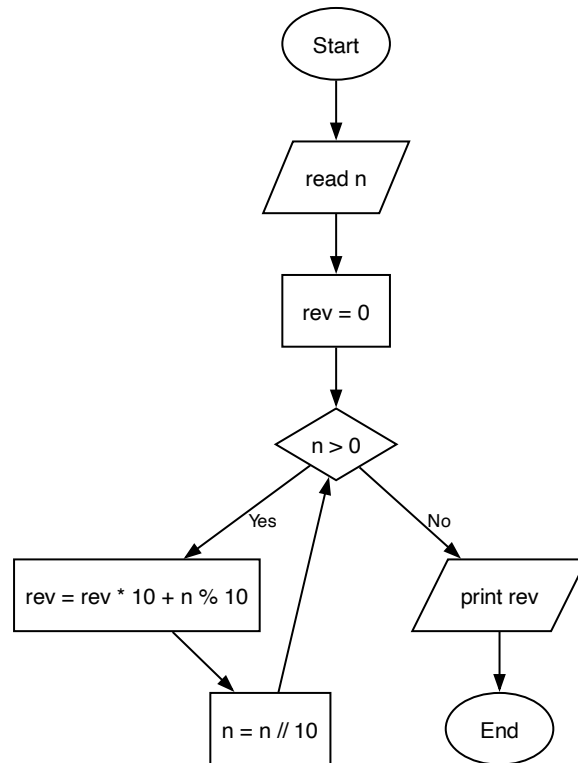
Trace this algorithm for $n = 3$, where the values of *marks* entered are 55, 82, 40 (in that order), and complete the trace table.



n	i	marks	output
3			
3	1		
3	1	55	
3	1	55	B
3	2	55	
3	2	82	
3	2	82	A
3	3	82	
3	3	40	
3	3	40	C
3	4	40	

Output: B, A, C

Problem 8.



(i) For each input below, follow the flowchart and write the output.

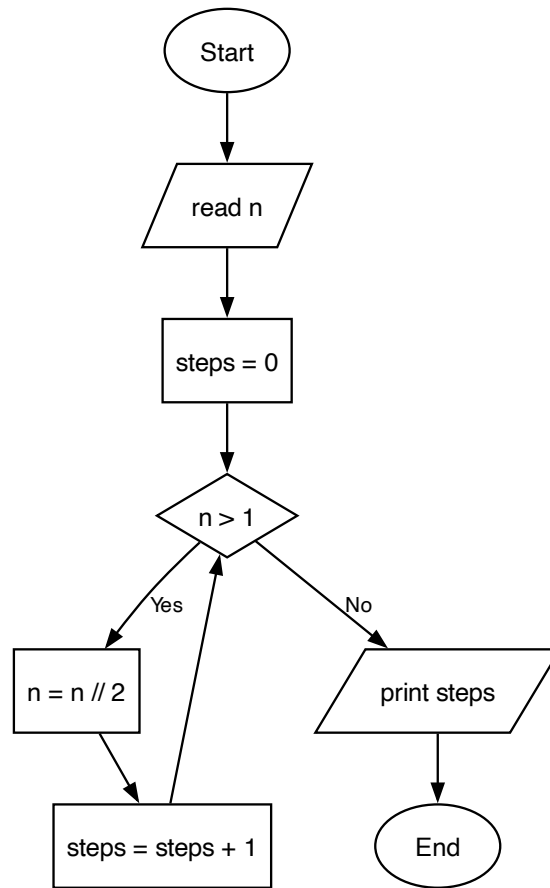
Note: $n \% 10$ gives the remainder after dividing by 10 — the last digit of n . For example, $472 \% 10 = 2$.

input n	output
123	321
47	74
5	5
470	74

(ii) Can you tell what this algorithm is doing?

Answer: Reverse the digits

Problem 9.



(i) For each input below, follow the flowchart and write the output.

input n	output
8	3
16	4
100	6
1000	9

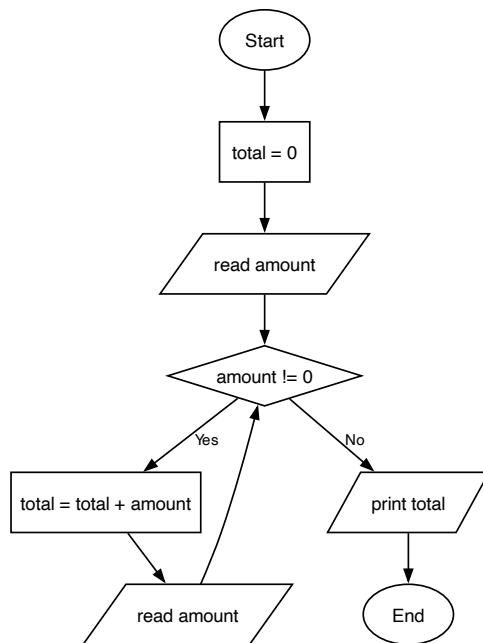
(ii) n grew from 8 all the way to 1000 — but look how little the output grew! Can you explain why?

Answer: Counting halvings — doubling n adds only ONE step. This is the secret behind binary search!

Problem 10.

A shopkeeper adds up the money received from each customer at the end of the day. She enters the amounts one by one, and enters 0 to say “no more” — the till then shows the total. Notice that the algorithm cannot know in advance how many customers there were!

Trace this algorithm, where the values of *amount* entered are 120, 75, 200, 0 (in that order), and complete the trace table.



total	amount	output
0		
0	120	
120	120	
120	75	
195	75	
195	200	
395	200	
395	0	
395	0	395

Output: 395